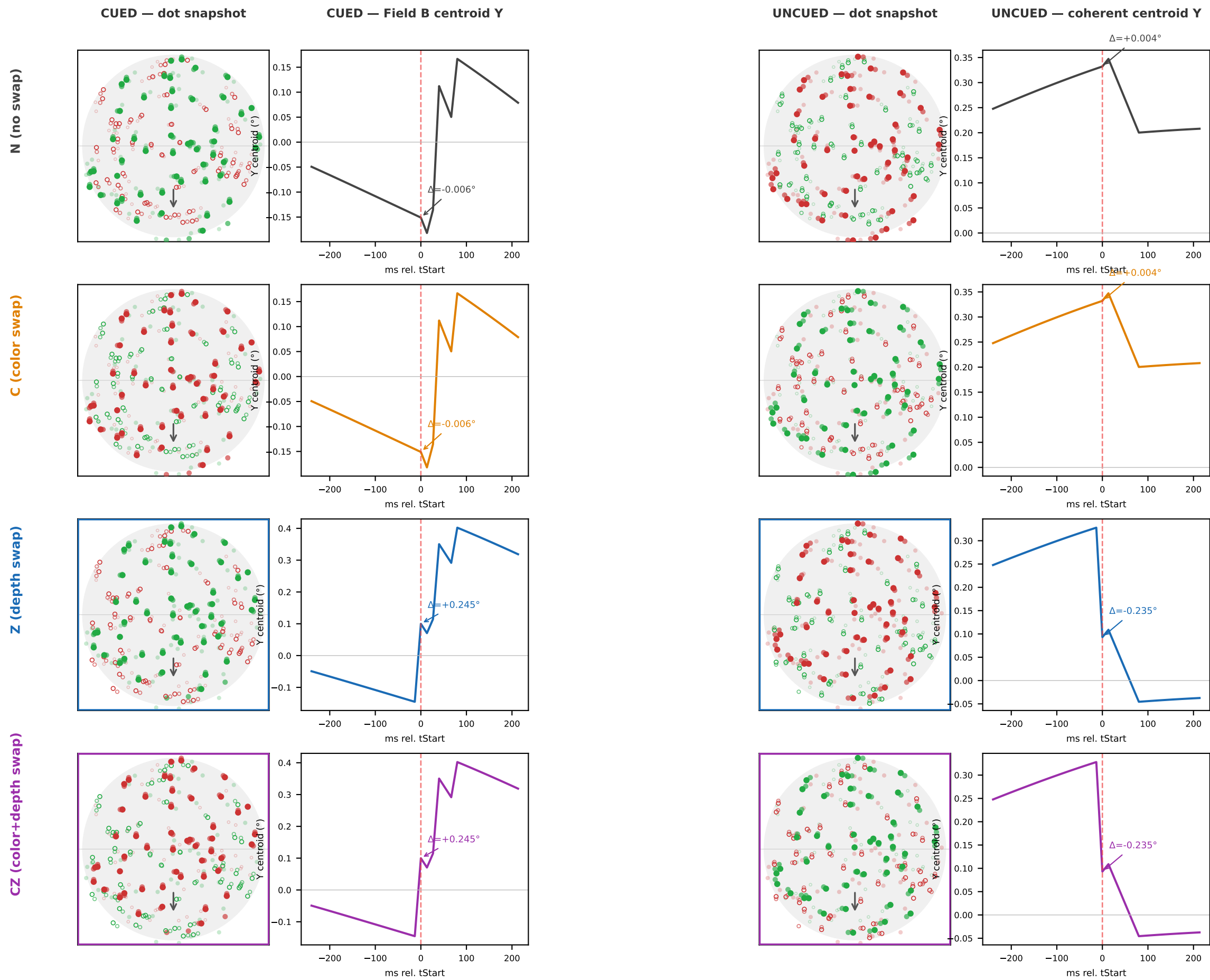


DecoupledDots — STEREO TRACES (PRE-FIX, buggy code)

Depth offset applied along transform.forward pitched 5.0° above gaze. Artifact visible as upward jump in Z and CZ centroid strips.

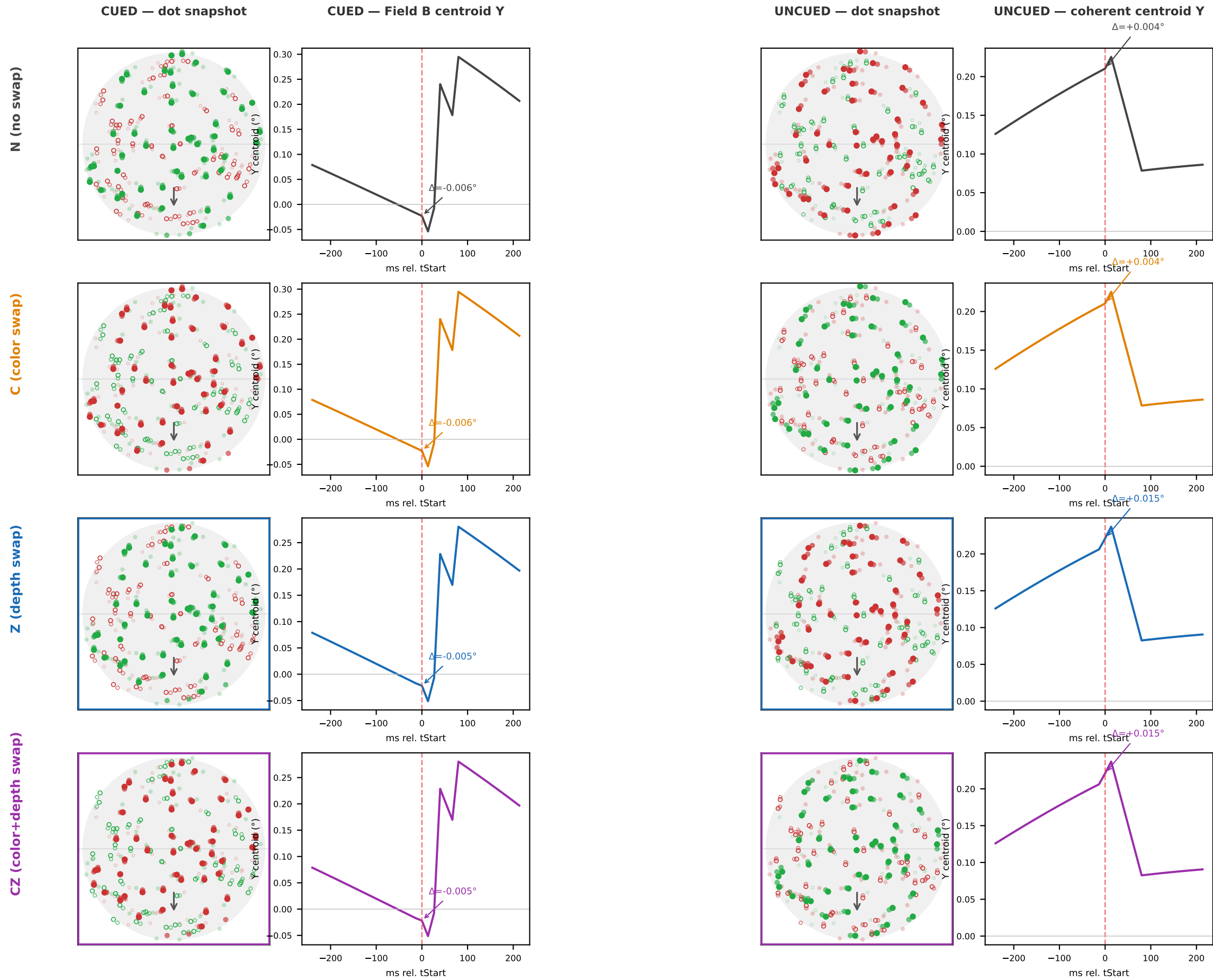


Heading = 270° (DOWN) throughout. Dots shown at frames tStart - 6 (faint), tStart (medium), tStart + 2 (solid). Colored border = depth swap active.

Red field Green field tStart Filled = Field B (delayed onset)

DecoupledDots — STEREO TRACES (POST-FIX, camera.forward)

Depth offset applied along camera.forward (gaze-aligned). No directional artifact. Tiny radial expansion at tStart in Z/CZ is sub-threshold.



Heading = 270° (DOWN) throughout. Dots shown at frames tStart - 6 (faint), tStart (medium), tStart + 2 (solid). Colored border = depth swap active.

Red field Green field - - - tStart Filled = Field B (delayed onset)

Centroid Y-jump at tStart: pre-fix vs post-fix

Coherent subfield (Field B, CUED arm), heading = 270° (DOWN)

Values in degrees of visual angle. Translation signal = 2.26 deg/sec × 13ms/frame = 0.030°/frame.

Condition	Pre-fix jump (°)	Post-fix jump (°)	Artifact/signal ratio (pre)	Data: % UP in wrong resp.
N (no swap)	-0.0057°	-0.0057°	0.2×	4%
C (color swap)	-0.0057°	-0.0057°	0.2×	6%
Z (depth swap)	+0.2453°	-0.0046°	8.1×	50%
CZ (color+depth swap)	+0.2453°	-0.0046°	8.1×	51%

Summary:

- N and C conditions: zero artifact in both pre-fix and post-fix (no depth swap at tStart).
- Z and CZ conditions: pre-fix jump is 8–9× the per-frame translation signal → dominates direction percept.
- Post-fix: residual jump ≈ 0.002° (perspective-induced radial expansion) — < 0.1× signal.
- The pre-fix artifact direction is always upward (+Y) regardless of true translation direction, consistent with observer GS reports of "jerky upward motion" and the UP-bias in wrong responses.
- After fix: stereo traces for Z and CZ look identical to N (no directional jump at tStart).